

Third, consider $E(\mathbf{V})/E(\mathbf{R})$. When all hypotheses are true it is identically 1, and again impossible to control. A remedy may be given by either adding 1 to the denominator, a somewhat artificial solution, or by changing the denominator to $E(\mathbf{R}|\mathbf{R} > 0)$. Modifying both numerator and denominator in the same way will again run into problems of control when $m_0 = m$.

3. FALSE DISCOVERY RATE CONTROLLING PROCEDURE

3.1. The Procedure

Consider testing $H_1, H_2, \dots, H_m$ based on the corresponding p -values $P_1, P_2, \dots, P_m$. Let $P_{(1)} \leq P_{(2)} \leq \dots \leq P_{(m)}$ be the ordered p -values, and denote by $H_{(i)}$ the null hypothesis corresponding to $P_{(i)}$. Define the following Bonferroni-type multiple-testing procedure:

let k be the largest i for which $P_{(i)} \leq \frac{i}{m} q^*$;

then reject all $H_{(i)}$ $i = 1, 2, \dots, k$. (1)

Theorem 1. For independent test statistics and for any configuration of false null hypotheses, the above procedure controls the FDR at q^* .

Proof. The theorem follows from the following lemma, whose proof is given in Appendix A.

Lemma. For any $0 \leq m_0 \leq m$ independent p -values corresponding to true null hypotheses, and for any values that the $m_1 = m - m_0$ p -values corresponding to the false null hypotheses can take, the multiple-testing procedure defined by procedure (1) above satisfies the inequality

$$E(\mathbf{Q} | P_{m_0+1} = p_1, \dots, P_m = p_{m_1}) \leq \frac{m_0}{m} q^*. \quad (2)$$

Now, suppose that $m_1 = m - m_0$ of the hypotheses are false. Whatever the joint distribution of $P_1'', \dots, P_{m_1}''$ which corresponds to these false hypotheses is, integrating inequality (2) above we obtain

$$E(\mathbf{Q}) \leq \frac{m_0}{m} q^* \leq q^*,$$

and the FDR is controlled.

Remark. Note that the independence of the test statistics corresponding to the false null hypotheses is not needed for the proof of the theorem.

This procedure was mentioned by Simes (1986) as an exploratory extension to his procedure for rejection of the intersection hypothesis that all null hypotheses are true if, for some i , $P_{(i)} \leq i\alpha/m$. Whereas Simes (1986) showed that his procedure controls the FWER under the intersection null hypothesis, Hommel (1988) showed that the extended procedure for inference on individual hypotheses does not control the FWER in the strong sense: for some configuration of the false null hypotheses, the probability of an erroneous rejection is greater than α .

Hochberg (1988) has suggested a different way to utilize Simes's procedure so that it does control the FWER in the strong sense, by offering the following procedure: